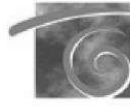


المادة: الفيزياء الشهادة: الثانوية العامة الفرع: علوم الحياة نموذج رقم 2 المدة: ساعتان	الهيئة الأكاديمية المشتركة قسم: العلوم	 المركز التربوي للبحوث والإنماء
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نموذج مسابقة (يراعي تعليق الدروس والتوصيف المعدل للعام الدراسي 2016-2017 وحتى صدور المناهج المطوّرة)
المرجع: دورة سنة 2012 الإستثنائية الإكمالية (معدّلة بحسب توصيف مادة الفيزياء للعام الدراسي 2016-2017)

This test includes three mandatory exercises. The use of non-programmable calculators is allowed.

Exercise 1 (7 points) Effect of the frequency on the current

The circuit, represented in the adjacent document (Doc 1), includes in series:

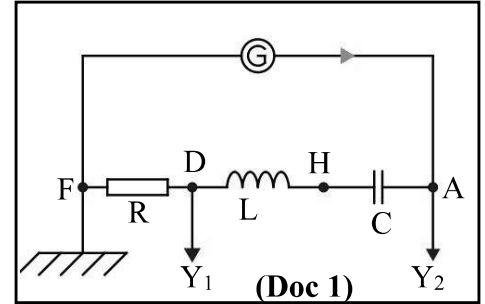
- A generator (G) delivering, across its terminals, an alternating voltage, $u_{AF} = u_G = 8\sin(2\pi ft)$ (S.I.);
- A capacitor of capacitance $C = 0.265 \mu\text{F}$;
- A coil of inductance $L = 31.833 \text{ mH}$ and of negligible resistance;
- A resistor of resistance $R = 100 \Omega$.

The circuit carries then an alternating current i of expression:

$$i = I_m \sin(2\pi ft + \varphi) \text{ (S.I.)}$$

The aim of this exercise is to study the effect of the frequency f of u_G on the amplitude I_m of i and on the phase difference φ between i and u_G .

An oscilloscope, connected as shown in the document (Doc.1), is used to display the voltages u_G and $u_R = u_{DF}$. The vertical sensitivity, of both channels, is the same in all the experiments: $S_V = 2 \text{ V/div}$.



1) 1st experiment

We set the frequency at $f = f_1 = 1500 \text{ Hz}$. We observe on the screen of the oscilloscope the waveforms displayed in the adjacent document (Doc.2).

1-1) Identify the waveforms (a) and (b).

1-2) Determine the phase difference φ_1 between i and u_G .

1-3) Calculate the amplitude I_{1m} of the current i .

2) 2nd experiment.

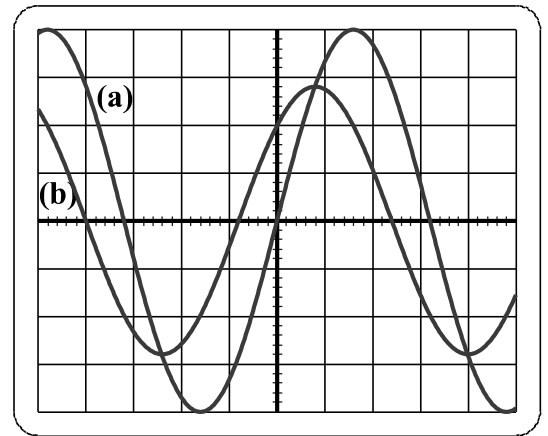
The frequency f is increased to $f = f_0$, f_0 being the proper frequency of the (RLC) series circuit.

We notice that the waveforms obtained coincide. The circuit is thus the seat of a certain phenomenon.

2-1) Give the name of the physical phenomenon obtained.

2-2) Give the value of the new phase difference φ_2 between i and u_G .

2-3) Deduce the value of f_0 and the new amplitude I_{2m} of i .



(Doc 2)

3) 3rd experiment

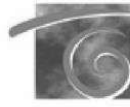
3-1) We measure I_m and φ for three other values of f ; the results are tabulated as shown in the adjacent table (Doc 3). Complete this table.

f (Hz)	1000	1500	$f_0 = ?$	2220	2500
I_m (A)	0.02			0.04	0.03
φ (rd)	1.33			-1.04	-1.2

(Doc 3)

3-2) Referring to the table (Doc 3), draw the graph representing the variation of I_m as a function of f .

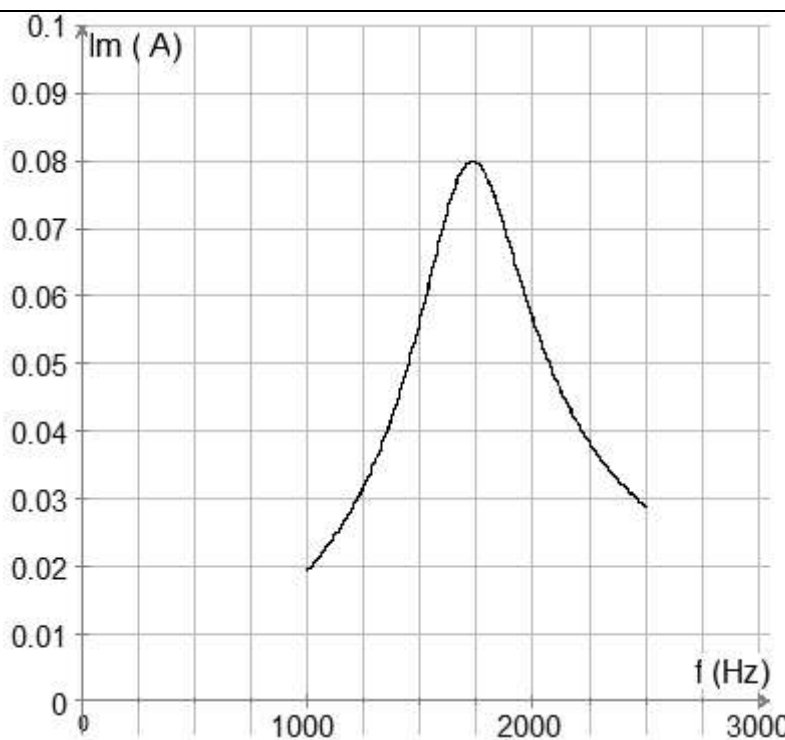
3-3) Conclude about the effect of f on the amplitude I_m of i and on the sign of the phase difference φ between i and u_G .

المادة: الفيزياء الشهادة: الثانوية العامة الفرع: علوم الحياة نموذج رقم 2 المدة: ساعتان	الهيئة الأكاديمية المشتركة قسم: العلوم	 المركز التربوي للبحوث والإنماء
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أسس التصحيح (تراعي تعليق الدروس والتوصيف المعدل للعام الدراسي 2016-2017 وحتى صدور المناهج المطورة)

Exercise 1 (7 points)

Effect of the frequency on the intensity of current

Question	Answer	Mark																		
1-1	$U_{mG} > U_{mR}$ with the same vertical sensitivity, (a) represents u_G and (b) represents u_R .	$\frac{1}{2}$																		
1-2	$ \varphi_1 = \frac{2\pi \times 0.8}{6.4} = \frac{\pi}{4}$ rd But the waveform (b) leads in phase the waveform (a), so u_R (or i) leads u_G because u_R reaches the maximum value before u_G , then $\varphi_1 = +\frac{\pi}{4}$ rd.	$\frac{1}{2}$																		
1-3	$I_{1m} = U_{Rm}/R = 0.056$ A	$\frac{1}{2}$																		
2-1	Current resonance.	$\frac{1}{4}$																		
2-2	$\varphi_2 = 0$	$\frac{1}{4}$																		
2-3	$LC\omega^2 = 1$ with $\omega = 2\pi f_0$, then $f_0 = \frac{1}{2\pi\sqrt{LC}} = 1733$ Hz. In case of current resonance, the circuit behaves as a pure resistor. So: $I_{2m} = U_{mG}/R = 8/100 = 0.08$ A	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$																		
3-1	<table><tr><td>f (Hz)</td><td>1000</td><td>1500</td><td>$f_0 = 1733$</td><td>2220</td><td>2500</td></tr><tr><td>I_m (A)</td><td>0.02</td><td>0.056</td><td>0.08</td><td>0.04</td><td>0.03</td></tr><tr><td>φ (rd)</td><td>1.33</td><td>0.785</td><td>0</td><td>-1.04</td><td>-1.2</td></tr></table>	f (Hz)	1000	1500	$f_0 = 1733$	2220	2500	I_m (A)	0.02	0.056	0.08	0.04	0.03	φ (rd)	1.33	0.785	0	-1.04	-1.2	$\frac{1}{2}$
f (Hz)	1000	1500	$f_0 = 1733$	2220	2500															
I_m (A)	0.02	0.056	0.08	0.04	0.03															
φ (rd)	1.33	0.785	0	-1.04	-1.2															
3-2		1																		
3-3	When f increases, for $f < f_0$, I_m increases and i leads u_G in phase; $\varphi > 0$. For $f = f_0$, I_m takes a maximum value and i and u_G are in phase; $\varphi = 0$. When f increases, for $f > f_0$, I_m decreases and i lags behind u_G in phase; $\varphi < 0$.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$																		

Exercise 2 (7 points)

Determination of the characteristics of electric components

The aim of this exercise is to determine the characteristics R , L and C respectively of a resistor, a coil of negligible resistance and a capacitor. For this, we perform two experiments. Take: $\pi^2 = 10$.

1) 1st experiment

Consider a series circuit (Doc 2) that consists of an LFG which delivers across its terminals an alternating sinusoidal voltage of effective value U and of adjustable frequency f , a resistor of resistance R , a coil of inductance L and of negligible resistance, a capacitor of capacitance C and an ammeter.

A voltmeter, connected across the terminals of the LFG, reads a constant value of $U = 21$ V.

We give f different values and we register, for each value, the effective current carried by the circuit. We obtain the plotted graph of (Doc 3) giving the variations of I as a function of f .

1-1) Specify the name of the physical phenomenon that takes place for $f = 200$ Hz.

1-2) Indicate then the proper frequency f_0 of this circuit.

1-3) Deduce the value of R .

1-4) Show that the first relation between L and C is:

$$LC = 0.625 \times 10^{-6} \text{ SI.}$$

2) 2nd experiment

We consider the RLC series circuit shown in (Doc 4) where $R = 150 \Omega$.

The expression of the voltage across the terminals of the LFG is: $u_{AM} = U_m \sin(2\pi ft)$.

The circuit thus carries an alternating sinusoidal current i .

The oscilloscope is connected to display the voltage u_{AM} across the LFG and the voltage u_{DM} across the resistor. (Doc 5) shows the waveforms (1) and (2) corresponding respectively to the voltages u_{AM} and u_{DM} , the frequency of u_{AM} being adjusted to $f = 50$ Hz.

The vertical sensitivity on both channels is 5 V/division.

2-1) Calculate, referring to (Doc 5), the maximum voltage U_m across the LFG.

2-2) Determine, referring to (Doc 5), the expression of the voltage u_{DM} .

2-3) Deduce the expression of i .

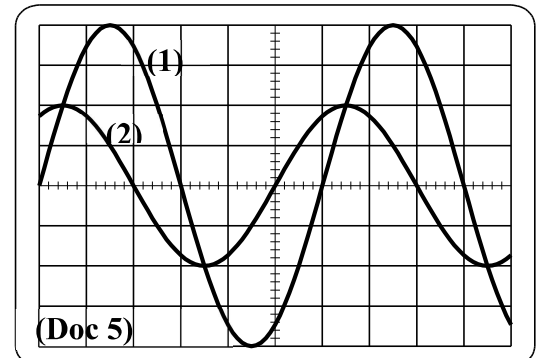
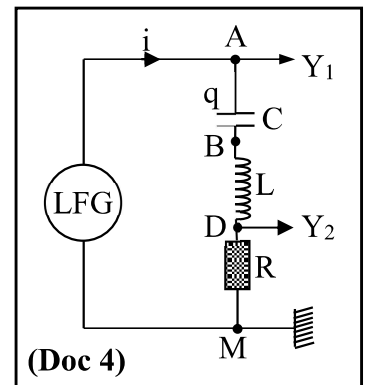
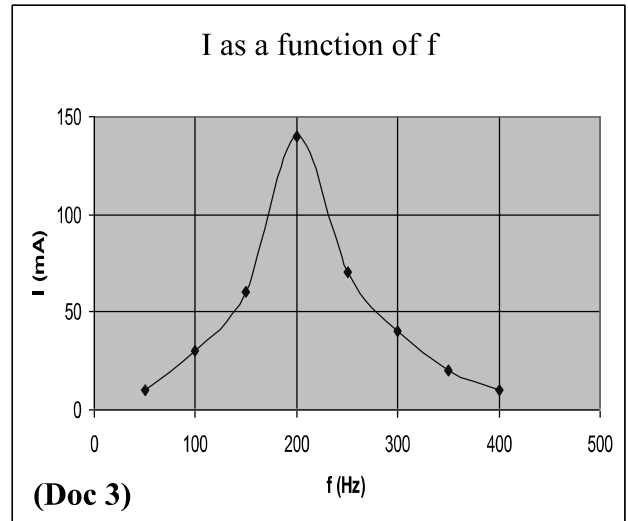
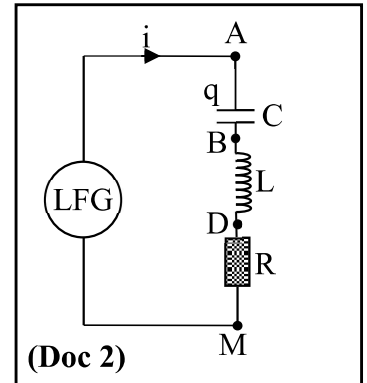
2-4) Determine the expression of the voltage u_{AB} across the terminals of the capacitor.

2-5) Determine the expression of the voltage u_{BD} across the terminals of the coil.

2-6) Using the relation $u_{AM} = u_{AB} + u_{BD} + u_{DM}$, at any instant t , and giving t the value zero ($t = 0$), show that the second relation between L and C is: $10^4 \pi^2 LC + 15000 \pi C \sqrt{3} = 1$.

3) Conclusion

Determine the values of L and C from the above two relations between L and C .



Exercise 2 (7 points)
Determination of the characteristics of electric components

Question	Answer	Mark
1-1	The circuit is thus the seat of the current resonance phenomenon since the effective current takes a maximum value I_0 for $f = 200$ Hz.	$\frac{1}{4}$
1-2	The proper frequency is then $f_0 = 200$ Hz.	$\frac{1}{4}$
1-3	The maximum value of the effective current is: $I_0 = 140$ mA. So: $R = \frac{U}{I_0} = \frac{21}{0.140} = 150 \Omega$.	$\frac{1}{2}$
1-4	In this case: $f_0 = \frac{1}{2\pi\sqrt{LC}} = 200$; $LC = \frac{1}{4 \times \pi^2 \times 4 \times 10^4}$ $LC = 0.625 \times 10^{-6}$ SI (1)	$\frac{1}{2}$
2-1	$U_m = S_v \cdot Y = 4 \times 5 = 20$ V $u_{AM} = 20 \sin(100\pi t)$	$\frac{1}{2}$
2-2	The waveform (2), (u_{DM}), leads in phase the waveform (1), (u_{AM}), by $ \varphi $. One period (2π) extends over 6 div; the phase difference $ \varphi $ is relative to 1 div. So: $ \varphi = \frac{2\pi \times 1}{6} = \frac{\pi}{3}$ rad and $\omega = 2\pi f = 100\pi$ rad/s $U_{m2} = S_v \cdot Y = 2 \times 5 = 10$ V and $u_{DM} = 10 \sin(100\pi t + \pi/3)$ (u_{DM} in V, t in s)	$1\frac{1}{2}$
2-3	Ohm's law gives: $i = \frac{u_{DM}}{R} = \frac{10}{150} \sin(100\pi t + \frac{\pi}{3})$; We get: $i = \frac{1}{15} \sin(100\pi t + \frac{\pi}{3}) = 0.067 \sin(100\pi t + \frac{\pi}{3})$; (i in A, t in s).	$\frac{1}{2}$
2-4	$i = \frac{dq}{dt} = C \frac{du_{AB}}{dt}$ The voltage across the capacitor is written as: $u_{AB} = \frac{1}{C} \int i dt = -\frac{1}{1500\pi C} \cos(100\pi t + \frac{\pi}{3})$, the integration constant being nil since u_{AB} is an alternating sinusoidal voltage.	$\frac{1}{2}$
2-5	$u_{BD} = L \frac{di}{dt} = \frac{100}{15} \pi L \cos(100\pi t + \frac{\pi}{3})$	$\frac{1}{2}$
2-6	$u_{AM} = u_{AB} + u_{BD} + u_{DM} \quad \forall t$ $20 \sin(100\pi t) = (\frac{100}{15} \pi L - \frac{1}{1500\pi C}) \cos(100\pi t + \frac{\pi}{3}) + 10 \sin(100\pi t + \frac{\pi}{3})$ For $t = 0$ $0 = (\frac{100}{15} \pi L - \frac{1}{1500\pi C}) \cos(\frac{\pi}{3}) + 10 \sin(\frac{\pi}{3})$ $10^4 \pi^2 LC + 15000\pi C \sqrt{3} = 1$ (2)	1
3	The equations (1) and (2) give: $\begin{cases} C = 1.15 \times 10^{-5} \text{ F} = 0.115 \mu\text{F} \\ L = 0.0543 \text{ H} = 54.3 \text{ mH} \end{cases}$	1